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Apparatus for inputting signal from communication line and method thereof

Abstract

Provided is an apparatus including: a first input circuit which inputs signal from a first communication line; a second input circuit which inputs a signal from a second communication line; a reference voltage generation circuit which generates a reference voltage; and a comparison circuit which compares the reference voltage with a voltage of the first communication line. The first input circuit and the second input circuit adjust judgement thresholds to be used for judging a signal value according to the comparison result obtained by the comparison circuit.

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Background/Summary

(1) The contents of the following patent application(s) are incorporated herein by reference: NO. 2022-209279 filed in JP on Dec. 27, 2022

BACKGROUND

1. Technical Field

(2) The present invention relates to an apparatus which inputs a signal from a communication line, and a method thereof.

2. Related Art

(3) Conventionally, there is known a serial bus interface such as I.sup.2C. Patent Document 1 describes that: a I.sup.2C serial communication bus uses a full-duplex serial line composed of two lines which are a serial data line (SDA) and a serial clock line (SCL); the SDA line and the SCL line are each connected to Vdd through a pull-up resistor; the Vdd is usually from 1.2 V to 3.3 V; and so on (refer to BACKGROUND OF THE INVENTION).

(4) Also, Patent Document 1 describes that: a voltage Vcap that approximates a voltage of the Vdd is generated by charging a capacitor C1 with a voltage corresponding to an SDA signal and a SCL signal; the voltage Vcap provides a reference voltage for a Schmitt trigger 208a and a Schmitt trigger 208b; and so on (refer to columns 4 to 5).

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: Specification of U.S. Pat. No. 8,698,543

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates examples of voltage specifications for a communication line.
- (2) FIG. 2 illustrates structure of an apparatus 2 according to the present embodiment.
- (3) FIG. 3 illustrates an operational flow of the apparatus 2 according to the present embodiment.
- (4) FIG. 4 illustrates operational timings of the apparatus 2 according to the present embodiment.
- (5) FIG. 5 illustrates structure of an apparatus 502 according to a variant example.
- (6) FIG. 6 illustrates structure of an obtainment circuit 640 according to the variant example.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(7) Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to claims. In addition, not all of the combinations of features described in the embodiments are essential to the solution of the invention.

(8) FIG. 1 illustrates examples of voltage specifications for a communication line. The examples in the present drawing show voltage specifications of a signal that is transmitted by means of SDA and SCL communication lines of an I^{sup}.2C interface. In these voltage specifications, an operation voltage V_{sub}.DD is defined by three types of standards which are: a 1.2 V standard (with a typical value of 1.2 V in the drawing); a 1.8 V standard (with a typical value of 1.8 V in the drawing); and a 3.3 V standard (with a typical value of 3.3 V in the drawing). Both of an allowable minimum voltage and an allowable maximum voltage of V_{sub}.DD are defined for the 1.2 V standard, 1.8 V standard, and 3.3 V standard. For example, for the 1.2 V standard, the V_{sub}.DD should be within a range from 1.10 V to 1.30 V.

(9) A LOW level input voltage V_{sub}.IL needs to be within a range from $-0.1 \times V_{sub}.DD$ to $0.3 \times V_{sub}.DD$. A HIGH level input voltage V_{sub}.IH needs to be within a range from $0.7 \times V_{sub}.DD$ to $1.1 \times V_{sub}.DD$. An input hysteresis V_{sub}.hys of a Schmitt trigger to be used for signal input should be $0.1 \times V_{sub}.DD$ or more.

(10) While an I^{sup}.2C interface having a constant voltage has been standardized where 1.8 V standard, 2.0 V standard, etc. are used, with a progress in low-voltage standardization in recent years, a 1.2 V standard has also been used for an I^{sup}.2C interface. Therefore, for systems which perform device-to-device communication by means of the I^{sup}.2C interface, there are a system using V_{sub}.DD of the 1.8 V standard, and a system using V_{sub}.DD of the 1.2 V standard. Because of this, it is desirable for apparatuses to be used for such systems, such as IC devices or LSI devices, to support both of a 1.8 V standard I^{sup}.2C interface and a 1.2 V standard I^{sup}.2C interface.

(11) When supporting both of the 1.8 V standard I^{sup}.2C interface and the 1.2 V standard I^{sup}.2C interface, a minimum value of a HIGH level input voltage V_{sub}.IH (minimum voltage when the SDA and the SCL are at logic H) is calculated as 0.7×1.1 V (minimum value of the V_{sub}.DD)=0.77 V. A maximum value of a LOW level input voltage V_{sub}.IL (maximum voltage when the SDA and the SCL are at logic L) is calculated as 0.3×1.9 V (maximum value of the V_{sub}.DD)=0.57 V. Here, if supporting the both of the 1.8 V standard and the 1.2 V standard with an input buffer having one input hysteresis, an input hysteresis V_{sub}.hys of the input buffer is calculated as 0.1×1.9 V (maximum value of the V_{sub}.DD)=0.19 V.

(12) Therefore, if supporting the both of the 1.8 V standard and the 1.2 V standard with the input buffer having one threshold value and the one input hysteresis, the input hysteresis V_{sub}.hys needs to be 0.19 V in spite of a fact that there is a margin of only 0.2 V (=0.77 V−0.57 V) between the minimum value 0.77 V of the HIGH level input voltage V_{sub}.IH and the maximum value 0.57 V of the LOW level input voltage V_{sub}.IL. Hence, it is difficult to support the 1.8 V standard and the 1.2 V standard by means of such an input buffer.

(13) FIG. 2 illustrates structure of an apparatus 2 according to the present embodiment. For example, the apparatus 2 is an IC device, an LSI device, or a multi-chip module (MCM). The apparatus 2 includes a communication interface for communicating with another IC device, or the like. The apparatus 2 according to the present embodiment supports, as an example of such a communication interface, an I.sup.2C interface that is a serial communication interface, a successor interface of the I.sup.2C interface, or the like. The apparatus 2 includes an external terminal P_SCL (also referred to as a “first external terminal”) connected to a SCL communication line (also referred to as a “first communication line”) which serves as a clock communication line of the I.sup.2C interface, and an external terminal P_SDA (also referred to as a “second external terminal”) connected to a SDA communication line (also referred to as a “second communication line”) which serves as a data communication line of the I.sup.2C interface.

(14) For convenience of explanation, the present drawing shows a structural part related to a receiver which receives signals from the first external terminal and the second external terminal in the apparatus 2. The apparatus 2 may further include a transmitter which transmits signals to another apparatus through the first external terminal and the second external terminal.

(15) The apparatus 2 may be a four-terminal device including a power supply terminal and a ground terminal, in addition to the first external terminal and the second external terminal. Such an apparatus 2 may be a sensor device such as a magnetic sensor, and supplies another apparatus with a measurement value through the communication interface. Instead of this, the apparatus 2 may include more terminals, and may also include another communication interface for communicating with another apparatus, apart from the communication interface using the first external terminal and the second external terminal.

(16) The apparatus 2 according to the present embodiment adjusts a judgement threshold of an input circuit, based on a result obtained by comparing a voltage of a communication line with a reference voltage. This enables the apparatus 2 to be adaptable to voltage specifications of multiple types. By way of example, the apparatus 2 according to the present embodiment supports two types of voltage specifications being 1.8 V standard and 1.2 V standard in an I.sup.2C interface. The apparatus 2 includes a first input circuit 200, a second input circuit 210, a reference voltage generation circuit 220, a comparison circuit 230, and an obtainment circuit 240.

(17) The first input circuit 200 is connected to the first external terminal, and inputs a signal from the first communication line. Note that, in the present specification, a word “connected” used hereinafter means “electrically connected” unless specifically noted, and its meaning is not only limited to a direct electrical connection between components but also an indirect electrical connection including another component connected between components. The first input circuit 200 outputs a logical value corresponding to a voltage value of the signal from the first communication line as a signal value SCLIN from the first communication line, to another internal circuit (unillustrated) in the apparatus 2. The first input circuit 200 has an input detector 204, input detectors 206a to 206b, and a selector 208.

(18) The input detector 204 is connected to the first external terminal. For example, the input detector 204 may be an input buffer circuit, and detects whether a signal value of the first communication line is at a HIGH level. Here, the input detector 204 detects whether a voltage of the first communication line exceeds a threshold voltage (also referred to as a “first voltage”). This first voltage is set to such a value by which the signal value of the first communication line can be detected as being at the HIGH level in any voltage specification among the voltage specifications of the multiple types. In the present embodiment, the input detector 204 has the first voltage as the threshold voltage which exceeds a maximum value of a LOW level voltage ($0.3 \times 1.9 = 0.57$ V) in the 1.8 V standard and is less than a minimum value of a HIGH level voltage ($0.7 \times 1.10 = 0.77$ V) in the 1.2 V standard. The input detector 204 may not have an input hysteresis, or may have an input hysteresis less than an input hysteresis defined in the communication interface.

(19) The multiple input detectors 206a to 206b (also referred to as “input detectors 206”) are

connected to the first external terminal. The multiple input detectors **206** may each be an input buffer circuit, for example. The multiple input detectors **206** detect signal values corresponding to the voltage of the first communication line by using judgement thresholds different from each other. Here, the multiple input detectors **206** may each be provided corresponding to each voltage specification of the voltage specifications of the multiple types, and has a judgement threshold for a corresponding voltage specification.

(20) In the present embodiment, the input detector **206a** is used for the 1.8 V standard, and has the judgement threshold that exceeds the maximum value of the LOW level voltage (e.g., $0.3 \times 1.9 = 0.57$ V) and is less than the minimum value of the HIGH level voltage (e.g., $0.7 \times 1.7 = 1.19$ V). The input detector **206b** is used for the 1.2 V standard, and has the judgement threshold that exceeds the maximum value of the LOW level voltage (e.g., $0.3 \times 1.30 = 0.39$ V) and is less than the minimum value of the HIGH level voltage (e.g., $0.7 \times 1.10 = 0.77$ V). The input detectors **206a** to **206b** have input hystereses satisfying the voltage specifications of the communication interface.

(21) The selector **208** is connected to the obtainment circuit **240** and the multiple input detectors **206**. The selector **208** selects a signal value output by one input detector **206** between the multiple input detectors **206** as output of the first input circuit **200**, depending on a comparison result obtained by the comparison circuit **230**. In the present embodiment, the selector **208** selects the input detector **206a** in response to receiving a comparison result indicating logic H from the obtainment circuit **240**, or selects the input detector **206b** in response to receiving a comparison result indicating logic L.

(22) The second input circuit **210** is connected to the second external terminal, and inputs a signal from the second communication line. The second input circuit **210** outputs a logical value corresponding to a voltage value of the signal from the second communication line as a signal value SDAIN from the second communication line, to another circuit (unillustrated) in the apparatus **2**. The second input circuit **210** has an input detector **214**, input detectors **216a** to **216b**, and a selector **218**.

(23) The input detector **214** is connected to the second external terminal. For example, the input detector **214** may be an input buffer circuit, and detects whether the signal value of the second communication line is at a HIGH level. Here, the input detector **214** detects whether a voltage of the second communication line exceeds a threshold voltage (also referred to as a “second voltage”). This second voltage is set to such a value by which the signal value of the second communication line can be detected as being at a LOW level in any voltage specification among the voltage specifications of the multiple types. In the present embodiment, the input detector **214** has the second voltage as the threshold voltage which exceeds the maximum value of the LOW level voltage ($0.3 \times 1.9 = 0.57$ V) in the 1.8 V standard and is less than the minimum value of the HIGH level voltage ($0.7 \times 1.10 = 0.77$ V) in the 1.2 V standard. Note that, the second voltage may be equal to the first voltage, or may be different from the first voltage. The input detector **214** may not have an input hysteresis, or may have an input hysteresis less than an input hysteresis defined in the communication interface.

(24) The multiple input detectors **216a** to **216b** (also referred to as “input detectors **216**”) are connected to the second external terminal. The multiple input detectors **216** detect signal values corresponding to voltage of the second communication line by using judgement thresholds different from each other. Since the multiple input detectors **216** have functions and structure the same as those of the multiple input detectors **206**, explanations for them will be omitted except for the following differences.

(25) The selector **218** is connected to the obtainment circuit **240** and the multiple input detectors **216**. The selector **218** selects a signal value output by one input detector **216** between the multiple input detectors **216** as output of the second input circuit **210**, depending on a comparison result obtained by the comparison circuit **230**. In the present embodiment, the selector **218** selects the input detector **216a** in response to receiving a comparison result indicating logic H from the

obtainment circuit **240**, or selects the input detector **216b** in response to receiving a comparison result indicating logic L.

(26) The reference voltage generation circuit **220** generates the reference voltage. The reference voltage generation circuit **220** generates as the reference voltage, a threshold voltage for judging which voltage specification among the voltage specifications of the multiple types is operating the communication interface, based on the voltage of the first communication line. Here, the first communication line and the second communication line are pulled-up to a voltage (e.g. 1.8 V or 1.2 V) corresponding to the voltage specification of the communication interface, which results in having voltages approximately the same as the voltage corresponding to the voltage specification under a stable state at a HIGH level. Therefore, the reference voltage generation circuit **220** according to the present embodiment has the threshold voltage higher than 1.30 V that is the maximum voltage for the 1.2 V standard, and lower than 1.70 V (e.g. 1.4 V) that is the minimum voltage for the 1.8 V standard.

(27) The comparison circuit **230** is connected to the first external terminal and the reference voltage generation circuit **220**. The comparison circuit **230** compares the reference voltage with the voltage of the first communication line. In the present embodiment, the comparison circuit **230** outputs a comparison result indicating logic H (HIGH) when the voltage of the first communication line exceeds the reference voltage, and outputs a comparison result indicating logic L (LOW) when the voltage of the first communication line is the reference voltage or less.

(28) The obtainment circuit **240** is connected to the first input circuit **200**, the second input circuit **210**, and the comparison circuit **230**. The obtainment circuit **240** obtains the comparison result output by the comparison circuit **230** according to a predetermined timing. The obtainment circuit **240** has a timing detection circuit **244** and a storage circuit **248**.

(29) The timing detection circuit **244** is connected to the input detector **204** in the first input circuit **200**, and the input detector **214** in the second input circuit **210**. The timing detection circuit **244** determines a timing to obtain the comparison result output by the comparison circuit **230** by using the signal value of the first communication line detected by the input detector **204** and the signal value of the second communication line detected by the input detector **214**. The timing detection circuit **244** may determine as the comparison result obtainment timing, a timing at which a signal value of the first communication line is definitely at a HIGH level due to specification of the communication interface when considering relationship between the signal value of the first communication line and the signal value of the second communication line. The timing detection circuit **244** may determine such an obtainment timing by means of logic operations for the signal value of the first communication line detected by the input detector **204** and the signal value of the second communication line detected by the input detector **214**. A method for determining the obtainment timing performed by the obtainment circuit **240** will be described later.

(30) The storage circuit **248** is connected to the comparison circuit **230** and the timing detection circuit **244**. The storage circuit **248** may be a flip-flop, a register, a memory, etc. and obtains and stores the comparison result output by the comparison circuit **230** at the obtainment timing determined by the timing detection circuit **244**. In the example shown in the present drawing, the storage circuit **248** is a flip-flop, and performs sampling on the comparison result input into a D terminal in response to a rise in a timing signal from the timing detection circuit **244**, then outputs the comparison result from a Q terminal.

(31) This enables the first input circuit **200** and the second input circuit **210** to adjust the judgement thresholds to be used for judging signal values by switching input detectors in the first input circuit **200** and the second input circuit **210** depending on the comparison result obtained by the comparison circuit **230** and received through the obtainment circuit **240**. More specifically, when the voltage of the first communication line exceeds the reference voltage, the first input circuit **200** and the second input circuit **210** respectively select the input detector **206a** and the input detector **216a** which are for the voltage specification using a higher voltage (the 1.8 V standard in the

example of the present drawing) among the voltage specifications of the multiple types. When the voltage of the first communication line is the reference voltage or less, the first input circuit **200** and the second input circuit **210** respectively select the input detector **206b** and the input detector **216b** which are for the voltage specification using a lower voltage (1.2 V standard in the example of the present drawing) among the voltage specifications of the multiple types. Therefore, even when voltage specifications of multiple types cannot be supported by an input circuit having one threshold value, the apparatus **2** judges which voltage specification among the voltage specifications of multiple types is used for a communication interface and adjusts a judgement threshold of an input circuit, so that the apparatus **2** can support the voltage specification.

(32) In addition, the storage circuit **248** may store a comparison result availability flag indicating whether the comparison result is available, and output the comparison result availability flag to the first input circuit **200** and the second input circuit **210**. For example, after a power supply of the apparatus **2** being inserted, after the apparatus **2** being reset, or the like, the storage circuit **248** has a value indicating “unavailable” for the comparison result availability flag because comparison has never been performed by the comparison circuit **230**. The storage circuit **248** has a value indicating “available” for the comparison result availability flag in response to being instructed by the timing detection circuit **244** to obtain the comparison result obtained by the comparison circuit **230** (e.g., in response to receiving the obtainment timing from the timing detection circuit **244**).

(33) When the comparison result availability flag indicates “unavailable”, the selector **208** in the first input circuit **200** and the selector **218** in the second input circuit **210** respectively select output of the input detector **204** and output of the input detector **214**. When the comparison result availability flag indicates “available”, the selector **208** in the first input circuit **200** and the selector **218** in the second input circuit **210** respectively select output of the input detector **206a** and output of the input detector **216a**, or output of the input detector **206b** and output of the input detector **216b**, depending on the comparison result obtained by the comparison circuit **230**. This enables the apparatus **2** to use the signal values of the first external terminal and the second external terminal by means of the input detector **204** and the input detector **214**, in a state where it has not been judged which voltage specification among the voltage specifications of the multiple types will be used.

(34) The apparatus **2** may not only automatically selects a voltage specification of the communication interface by selecting output of the selector **208** and output of the selector **218** depending on the comparison result obtained by the comparison circuit **230**, but also make the automatic selection function for the voltage specification unavailable and set an unchanging voltage specification. For example, the apparatus **2** may set whether to make the automatic selection function available based on a specified voltage value for the communication interface, and a value of an automatic setting availability flag indicating whether the automatic selection function is available, and set which voltage specification to select when the automatic selection function is made unavailable.

(35) For example, when the automatic setting availability flag indicates “available”, as mentioned above, the selector **208** selects a signal value output by one input detector between the multiple input detectors **206** (or instead, the input detector **204**) as output of the first input circuit **200**, depending on the comparison result obtained by the comparison circuit **230** (the comparison result stored in the storage circuit **248**). Similarly, the selector **218** selects a signal value output by one input detector between the multiple input detectors **216** (or instead, the input detector **214**) as output of the second input circuit **210**.

(36) When the automatic setting availability flag indicates “unavailable”, the selector **208** selects a signal value output by one input detector between the multiple input detectors **206** (or instead, the input detector **204**) as output of the first input circuit **200**, according to the specified voltage value. Similarly, the selector **218** selects a signal value output by one input detector between the multiple input detectors **216** (or instead, the input detector **214**) as output of the second input circuit **210**.

Here, the specified voltage value may be a selection value which specifies either of the multiple input detectors **206** (or instead, the input detector **204**).

(37) The apparatus **2** may store the specified voltage value and the automatic setting availability flag in the storage circuit **248**, another internal register, or the like. The apparatus **2** may receive the specified voltage value and the automatic setting availability flag from another apparatus through the communication interface or another interface, and store the same inside the apparatus **2**. Note that, the apparatus **2** may set the automatic setting availability flag as “available” after the power supply of the apparatus **2** is inserted, after the apparatus **2** is reset, or the like.

(38) In this case, the another apparatus may read out the comparison result in the storage circuit **248** through the communication interface in response to the automatic setting for the voltage specification being completed in the apparatus **2**, and then the another apparatus may set the specified voltage value corresponding to the comparison result to the apparatus **2** as well as setting the automatic setting availability flag of the apparatus **2** as “unavailable”, through the communication interface. This enables the first input circuit **200** and the second input circuit **210** to continuously output unchanging output of the input detectors that is temporarily selected by means of the automatic selection function, from when the another apparatus performs the setting onward.

(39) Note that, if a voltage specification of the communication interface has already been known, the another apparatus may set a specified voltage value to the apparatus **2** as well as setting the automatic setting availability flag as “unavailable” through the communication interface, without reading out the comparison result stored in the storage circuit **248** of the apparatus **2**. This enables the apparatus **2** to correctly receive the specified voltage value through the communication interface of which the voltage specification has been set automatically.

(40) FIG. **3** illustrates an operational flow of the apparatus **2** according to the present embodiment. In **S300**, the first communication line and the second communication line have logic H signal values, and their voltages are at HIGH levels. For example, for the I.sup.2C interface, each communication line is pulled-up to a power source voltage V_{sub_DD} corresponding to a voltage specification (i.e. 1.8 V for the 1.8 V standard, 1.2 V for the 1.2 V standard) through a pull-up resistor. In a state in which none of apparatuses connected to the first communication line and the second communication line has started communication, all apparatuses release each of their communication lines, thus a voltage of the each of their communication lines rises to a power source voltage V_{sub_DD} .

(41) Because of this, a voltage of a signal from the first communication line exceeds the first voltage, the input detector **204** outputs logic H. Also, because a voltage of a signal from the second communication line exceeds the second voltage, the input detector **214** outputs logic H.

(42) In **S310**, the timing detection circuit **244** in the obtainment circuit **240** determines a timing to obtain the comparison result output by the comparison circuit **230** by using the signal value of the first communication line detected by the input detector **204** and the signal value of the second communication line detected by the input detector **214**. In the present embodiment, the timing detection circuit **244** waits until the signal from the second communication line changes by using the output of the input detector **214** (No in **S310**), and determines the comparison result obtainment timing in response to a timing of change in the signal from the second communication line (Yes in **S310**).

(43) In **S320**, the storage circuit **248** in the obtainment circuit **240** obtains the comparison result obtained by the comparison circuit **230** comparing the reference voltage with the voltage of the first communication line at the obtainment timing determined by the timing detection circuit **244**. Here, in the I.sup.2C interface, communication starts in response to a start condition being met by the logic H signal from the second communication line changing into a logic L signal during a period in which the signal from the first communication line is at logic H. Therefore, the timing of change for the logic H signal from the second communication line to change into the logic L signal during the period in which the signal from the first communication line is at logic H is definitely a timing

in which the signal value of the first communication line is at a HIGH level due to the specification of the I.sup.2C interface. Accordingly, the obtainment circuit **240** according to the present embodiment obtains the comparison result according to the timing of change in the signal from the second communication line during the period in which the voltage of the signal from the first communication line exceeds the first voltage.

(44) In **S330**, **S340**, and **S350**, the first input circuit **200** and the second input circuit **210** adjust judgement thresholds according to the comparison result obtained by the obtainment circuit **240**. When there is obtained the comparison result indicating that the voltage of the first communication line (the “SCL voltage” in the drawing) exceeds the reference voltage (e.g. 1.4 V) (i.e., Yes in **S330**), the selector **208** in the first input circuit **200** and the selector **218** in the second input circuit **210** respectively select signal values output by the input detector **206a** and the input detector **216a** (also referred to as “first input detectors”) as output of the first input circuit **200** and output of the second input circuit **210**. When there is obtained the comparison result indicating that the voltage of the first communication line is the reference voltage or less (i.e., No in **S330**), the selector **208** in the first input circuit **200** and the selector **218** in the second input circuit **210** respectively select the signal values output by the input detector **206b** and the input detector **216b** (second input detectors) as output of the first input circuit **200** and output of the second input circuit **210**. This enables the first input circuit **200** and the second input circuit **210** to adjust the judgement thresholds.

(45) In **S360**, the first input circuit **200** and the second input circuit **210** detect that a signal value of the second communication line is at logic L, and a logic H signal value of the first communication line has changed into a logic L signal value, by using the judgement thresholds after the adjustment (Yes in **S360**). From this, another circuit in the apparatus **2** which receives the signals SCLIN and SDAIN determines that the period for the start condition has ended. In **S370**, the first input circuit **200** and the second input circuit **210** respectively detect a signal value of the first communication line and a signal value of the second communication line by using the judgement thresholds after the adjustment, and the another circuit in the apparatus **2** receives data by using the signal SCLIN from the first input circuit **200** and the SDAIN from the second input circuit **210**.

(46) FIG. **4** illustrates operational timings of the apparatus **2** according to the present embodiment. Before time **t1**, a voltage of the first communication line exceeds the first voltage (logic H), and a voltage of the second communication line exceeds the second voltage (logic H), which corresponds to a state at **S300** in FIG. **3**. At time **t1**, once the start condition period starts in response to the signal from the second communication line being changed from logic H to logic L (which corresponds to Yes in **S310**), the obtainment circuit **240** obtains the comparison result obtained by the comparison circuit **230** (**S320**). The first input circuit **200** and the second input circuit **210** adjust the judgement thresholds according to the obtained comparison result (**S330**, **S340**, and **S350**).

(47) At time **t2**, once the logic H signal from the first communication line changes into a logic L signal, the start condition period ends (Yes in **S360**). At time **t2** onwards, data transfer using the first communication line and the second communication line is performed, in which the first input circuit **200** and the second input circuit **210** respectively receive a signal of the first communication line and a signal of the second communication line by using the adjusted judgement thresholds and output these signals as signals SCLIN and SDAIN to the another circuit in the apparatus **2**. At time **t3**, once the logic L signal from the second communication line rises to a logic H signal during the period in which the signal from the first communication line is at logic H, a stop condition is met and thus the data transfer ends.

(48) Subsequently, at time **t1'**, once a next start condition period starts in response to a logic H signal from the second communication line changing into a logic L signal, the obtainment circuit **240** obtains the comparison result obtained by the comparison circuit **230**. Then, the first input circuit **200** and the second input circuit **210** adjust the judgement thresholds according to the newly obtained comparison result.

(49) The obtainment circuit **240** according to the present embodiment obtains the comparison result obtained by the comparison circuit **230** according to a timing of change in the signal from the second communication line during the period in which a voltage of the signal from the first communication line exceeds the first voltage. This is not limited to the storage circuit **248** obtaining the comparison result obtained by the comparison circuit **230** simultaneously or approximately simultaneously with the timing of change in the signal from the second communication line. For example, the storage circuit **248** may obtain the comparison result at time **t2** after a predetermined delay time or before time **t2** according to the timing of change.

(50) In addition, the obtainment circuit **240** according to the present embodiment may obtain the comparison result within a period in which a voltage of a signal from the first communication line exceeds the first voltage and a voltage of a signal from the second communication line is the second voltage or less. For example, the obtainment circuit **240** may obtain the comparison result at any timing from time **t1** to time **t2** in the drawing. For example, the timing detection circuit **244** may output as the comparison result obtainment timing, a timing at which “(an output logical value of the input detector **204**) AND NOT (an output logical value of the input detector **214**)” rises. The timing detection circuit **244** may delay this timing of rise by a predetermined time and output the timing of rise as the comparison result obtainment timing.

(51) The obtainment circuit **240** can also obtain a comparison result corresponding to a voltage taken before the first communication line showing logic H falls to logic L or a voltage of the first communication line that is taken immediately before. For example, the timing detection circuit **244** may output as the comparison result obtainment timing, a timing at which “(an output logical value of the input detector **204**) AND NOT (an output logical value of the input detector **214**)” falls. The storage circuit **248** may obtain a value obtained by delaying the comparison result from the comparison circuit **230** by a predetermined delay time at the obtainment timing.

(52) Note that, the obtainment circuit **240** may determine an obtainment timing for obtaining the comparison result within a period in which the voltage of the signal from the first communication line exceeds the first voltage, according to a specification of the communication interface. For example, the obtainment circuit **240** may obtain the comparison result obtained by the comparison circuit **230** in response to a timing of change of the second communication line rising from logic L to logic H, depending on the specification of the communication interface.

(53) FIG. 5 illustrates structure of an apparatus **502** according to a variant example of the apparatus **2** in FIG. 2. In the present drawing, components having functions and structure the same as those of the apparatus **2** in FIG. 2 are denoted with the same reference numbers as those for the apparatus **2**, and explanation therefor will be omitted except for the following differences.

(54) The apparatus **502** includes an obtainment circuit **540** instead of the obtainment circuit **240**. In addition to the functions of the obtainment circuit **240**, the obtainment circuit **540** instructs to stop a comparison operation performed by a comparison circuit **230**, to stop signal detection operations performed by an input detector **204** and an input detector **214**, and to stop signal detection operations performed by input detectors **206a** to **206b** and input detectors **216a** to **216b**. The obtainment circuit **540** may have any one of, or two or more of these functions. The obtainment circuit **540** has a timing detection circuit **544** and a storage circuit **548**.

(55) The timing detection circuit **544** is connected to an input detector **204** in a first input circuit **200**, and an input detector **214** in a second input circuit **210**. The timing detection circuit **544** instructs the comparison circuit **230** to perform the comparison operation in a comparison period within a period in which a voltage of a signal from a first communication line exceeds a first voltage. The timing detection circuit **544** instructs the comparison circuit **230** to stop the comparison operation in a period other than the comparison period. The comparison circuit **230** executes/stops the comparison operation in response to the instruction from the timing detection circuit **544**.

(56) The timing detection circuit **544** may instruct the comparison circuit **230** to perform the

comparison operation in a comparison period being an entire period in which the voltage of the signal from the first communication line exceeds the first voltage, or a part of the entire period. Instead of this, the timing detection circuit **544** may instruct the comparison circuit **230** to perform the comparison operation in a comparison period being an entire period in which the voltage of the signal from the first communication line exceeds the first voltage and a voltage of a signal from a second communication line is a second voltage or less, or a part of the entire period. The timing detection circuit **544** may end the comparison period after the storage circuit **548** has finished obtaining the comparison result.

(57) While the comparison operation is stopped, the comparison circuit **230** may transition to a low-power consumption mode in which less power is consumed compared to a normal mode for executing the comparison operation. Instead of this, the comparison circuit **230** may be powered off while the comparison operation is stopped.

(58) Similar to the timing detection circuit **244**, the timing detection circuit **544** determines a timing to obtain the comparison result output by the comparison circuit **230** by using a signal value of the first communication line detected by the input detector **204** and a signal value of the second communication line detected by the input detector **214**. The timing detection circuit **544** may determine as the obtainment timing, a timing at which a value of the comparison result is stable after a predetermined margin time has been passed since a start of the comparison period.

(59) The storage circuit **548** is connected to the comparison circuit **230** and the timing detection circuit **544**. Similar to the storage circuit **248**, the storage circuit **548** obtains the comparison result output by the comparison circuit **230** at the obtainment timing determined by the timing detection circuit **244**, and stores the comparison result.

(60) The multiple input detectors **206** and the multiple input detectors **216** may switch whether to perform signal detection operation, depending on the comparison result obtained from the storage circuit **548**. At least one input detector **206** having output that is not selected by the selector **208** between the multiple input detectors **206**, and at least one input detector **216** having output that is not selected by the selector **218** between the multiple input detectors **216** may stop performing signal value detection operation. While the signal value detection operation is stopped, each of the input detectors **206** and each of the input detectors **216** may transition to low-power consumption modes in which less power is consumed compared to normal modes for executing the signal value detection operation. Instead of this, the each of the input detectors **206** and the each of the input detectors **216** may be powered off while the signal value detection operation is stopped.

(61) The input detector **204** and the input detector **214** may switch whether to perform the signal value detection operation, depending on the comparison result obtained from the storage circuit **548**. When the input detector **204** and the input detector **214** are respectively selected by the selector **208** and the selector **218**, they may execute the signal value detection operation, but when the input detector **204** and the input detector **214** are not respectively selected by the selector **208** and the selector **218**, they may stop performing the signal value detection operation. The input detector **204** and the input detector **214** may start the signal value detection operation in response to receiving a signal indicating a stop condition from a communication interface according to an instruction of the obtainment circuit **540**, and may stop performing the signal value detection operation after the timing for the storage circuit **548** to obtain the comparison result, or after a start condition obtained from the communication interface has ended.

(62) While the signal value detection operation is stopped, the input detector **204** and the input detector **214** may transition to low-power consumption modes in which less power is consumed compared to normal modes for executing the signal value detection operation. Instead of this, the input detector **204** and the input detector **214** may be powered off while the signal value detection operation is stopped.

(63) FIG. **6** illustrates structure of an obtainment circuit **640** according to the variant example. Because the obtainment circuit **640** is a variant example of the obtainment circuit **240** in FIG. **2**,

explanation thereof will be omitted except for the following differences.

(64) The obtainment circuit **640** has a storage circuit **648** and a reset circuit **644**. The storage circuit **648** is used instead of the storage circuit **248** in FIG. 2. By way of example, the storage circuit **648** is a set-reset flip-flop, which is set by a rise in a set input S so that an output Q is leveled at logic H, and is reset by a rise in a reset input R so that the output Q is leveled at logic L.

(65) The reset circuit **644** is used instead of the timing detection circuit **244** in FIG. 2. The reset circuit **644** inputs a signal that rises according to a timing of change in a signal from the second communication line during a period in which a voltage of a signal from the first communication line exceed a first voltage, to a reset input R of the storage circuit **648**. In this way, the storage circuit **648** resets the comparison result that has been obtained (previous comparison result) and set output at logic L. For example, the reset circuit **644** may be composed of a circuit which outputs a pulse with a predetermined time width according to a timing at which “(an output logical value of the input detector **204**) AND NOT (an output logical value of the input detector **214**)” rises.

(66) Here, the storage circuit **648** obtains a new comparison result indicating that a voltage of the first communication line has exceeded the reference voltage (comparison result indicating logic H), in response to the output of the comparison circuit **230** (i.e., set input S for the storage circuit **648**) rising after being reset. The storage circuit **648** does not obtain the new comparison result if the output of the comparison circuit **230** remains at logic L without rising after being reset, but as a result of that, the storage circuit **648** rather outputs a comparison result indicating that the voltage of the first communication line is the reference voltage or less (comparison result indicating logic L).

(67) According to the apparatus **2** and the apparatus **502** shown above, the judgement threshold of the input circuit can be adjusted according to a voltage specification being in use between the two types of the voltage specifications of the communication interface. In order to support three or more types of voltage specifications, the reference voltage generation circuit **220** may generate multiple threshold voltages corresponding to ranges of voltage specifications of multiple types. One or more comparison circuits **230** may use the multiple threshold voltages from the reference voltage generation circuit **220** to an output comparison result indicating that the voltage of the first communication line belongs to which one of multiple voltage ranges corresponding to the voltage specifications of the multiple types. The first input circuit **200** and the second input circuit **210** may respectively have the input detectors **206** and the input detectors **216** each of which corresponds to each of the voltage specifications of the multiple types. The selector **208** and the selector **218** may respectively select the input detector **206** and the input detector **216** to be used for judging a signal value depending on the comparison result obtained by the comparison circuit **230**.

(68) The apparatus **2** or the apparatus **502** shown above uses a serial interface as the communication interface. Instead of this, the apparatus **2** or the apparatus **502** may use a parallel interface as the communication interface. In this case, the communication interface may have one or two or more first communication lines, and two or more second communication lines. Due to specification of the communication interface, the apparatus **2** or the apparatus **502** may identify a period in which a voltage of the signal from the first communication line exceeds the first voltage based on signal values of the first communication line and the second communication line, and select the input detector **206** and the input detector **216** to be used for judging the signal value depending on the comparison result obtained by the comparison circuit **230** at this timing.

(69) While the present invention has been described by way of the embodiments, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above described embodiments. It is also apparent from description of the claims that embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

(70) It should be noted that the operations, procedures, steps, stages, and the like of each processing

performed by an apparatus, system, program, and method shown in the claims, specification, or drawings can be executed in any order as long as the order is not indicated by “before”, “prior to” or the like and as long as the output from previous processing is not used in later processing. Even if the operational flow is described using phrases such as “first” or “next” for the sake of convenience in the claims, specification, or drawings, it does not necessarily mean that the processing must be performed in this order.

EXPLANATION OF REFERENCES

(71) **2**: apparatus; **200**: first input circuit; **204**: input detector; **206a** to **206b**: input detectors; **208**: selector; **210**: second input circuit; **214**: input detector; **216a** to **216b**: input detectors; **218**: selector; **220**: reference voltage generation circuit; **230**: comparison circuit; **240**: obtainment circuit; **244**: timing detection circuit; **248**: storage circuit; **502**: apparatus; **540**: obtainment circuit; **544**: timing detection circuit; **548**: storage circuit; **640**: obtainment circuit; **644**: reset circuit; **648**: storage circuit.

Claims

1. An apparatus comprising: a first input circuit which inputs a signal from a first communication line in serial communication; a second input circuit which inputs a signal from a second communication line in the serial communication; a reference voltage generation circuit which generates a reference voltage; a comparison circuit which compares the reference voltage with a voltage of the first communication line; and an obtainment circuit which obtains a comparison result obtained by the comparison circuit according to a timing of change in the signal from the second communication line during a period in which a voltage of the signal from the first communication line exceeds a first voltage, wherein the first input circuit and the second input circuit adjust judgement thresholds to be used for judging signal values according to the comparison result obtained by the obtainment circuit.
2. The apparatus according to claim 1, wherein the obtainment circuit obtains the comparison result within a period in which the voltage of the signal from the first communication line exceeds the first voltage.
3. The apparatus according to claim 2, wherein the obtainment circuit obtains the comparison result within a period in which the voltage of the signal from the first communication line exceeds the first voltage, and a voltage of the signal from the second communication line is a second voltage or less.
4. The apparatus according to claim 3, wherein the obtainment circuit resets the comparison result that has been obtained, according to the timing of change in the signal from the second communication line during the period in which the voltage of the signal from the first communication line exceeds the first voltage, and obtains a new comparison result indicating that the voltage of the first communication line has exceeded the reference voltage in response to output of the comparison circuit rising after the resetting.
5. The apparatus according to claim 3, wherein input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on the comparison result obtained by the comparison circuit.
6. The apparatus according to claim 2, wherein input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a

selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on the comparison result obtained by the comparison circuit.

7. The apparatus according to claim 1, wherein input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on the comparison result obtained by the comparison circuit.

8. The apparatus according to claim 7, wherein at least one input detector, among the multiple input detectors, of which output was not selected by the selector stops performing signal value detection operation.

9. The apparatus according to claim 1, wherein the comparison circuit performs comparison operation in a comparison period within the period in which the voltage of the signal from the first communication line exceeds the first voltage, and stops performing the comparison operation in a period other than the comparison period.

10. The apparatus according to claim 9, wherein the comparison circuit performs the comparison operation within the comparison period in which the voltage of the signal from the first communication line exceeds the first voltage, and a voltage of the signal from the second communication line is a second voltage or less.

11. A method comprising: inputting, by a first input circuit, a signal from a first communication line in serial communication; inputting, by a second input circuit, a signal from a second communication line in the serial communication; generating, by a reference voltage generation circuit, a reference voltage; comparing, by a comparison circuit, the reference voltage with a voltage of the first communication line; obtaining, by an obtainment circuit, a comparison result obtained by the comparison circuit according to a timing of change in the signal from the second communication line during a period in which a voltage of the signal from the first communication line exceeds a first voltage; and adjusting, by the first input circuit and the second input circuit, judgement thresholds to be used for judging signal values according to the comparison result obtained by the obtainment circuit.

12. The method according to claim 11, wherein the obtainment circuit obtains the comparison result within the period in which the voltage of the signal from the first communication line exceeds the first voltage.

13. The method according to claim 12, wherein the obtainment circuit obtains the comparison result within a period in which the voltage of the signal from the first communication line exceeds the first voltage, and a voltage of the signal from the second communication line is a second voltage or less.

14. The method according to claim 13, wherein the obtainment circuit resets the comparison result that has been obtained, according to the timing of change in the signal from the second communication line during the period in which the voltage of the signal from the first communication line exceeds the first voltage, and obtains a new comparison result indicating that the voltage of the first communication line has exceeded the reference voltage in response to output of the comparison circuit rising after the resetting.

15. The method according to claim 13 wherein: input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on

the comparison result obtained by the comparison circuit.

16. The method according to claim 12, wherein: input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on the comparison result obtained by the comparison circuit.

17. The method according to claim 11, wherein: input circuits that are the first input circuit and the second input circuit, each has: multiple input detectors which detect, by using the judgement thresholds different from each other, signal values corresponding to a voltage of a corresponding communication line that is the first communication line or the second communication line; and a selector which selects a signal value output by one of the multiple input detectors as output of a corresponding input circuit that is the first input circuit or the second input circuit, depending on the comparison result obtained by the comparison circuit.

18. The method according to claim 17, wherein at least one input detector, among the multiple input detectors, of which output was not selected by the selector stops performing signal value detection operation.

19. The method according to claim 11, wherein the comparison circuit performs comparison operation in a comparison period within the period in which the voltage of the signal from the first communication line exceeds the first voltage, and stops performing the comparison operation in a period other than the comparison period.

20. The method according to claim 19, wherein the comparison circuit performs the comparison operation within the comparison period in which the voltage of the signal from the first communication line exceeds the first voltage, and a voltage of the signal from the second communication line is a second voltage or less.
